DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

2 TITLE	1 IDENTIFICATION NUMBER
EQUIPMENT AIRBORNE SOUND MEASUREMENT PLAN	DI-HFAC-80270

3 DESCRIPTION/PURPOSE

3.1 This plan provides a description of the airborne sound test procedure, facility and measurement system in advance of test to demonstrate planned compliance with requirements of MIL-STD-740-1.

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
861230	SH		

7. APPLICATION/INTERRELATIONSHIP

7.1 This Data Item Description (DID) contains the format and content preparation instructions for that data generated under the work task described by 4.2 of MIL-STD-740-1.

7.2 This DID may be applied for each grade, class, rating and type of equipemnt to be measured on each contract or order when MIL-STD-740-1 is invoked.

7.3 The data generated under the requirements of this DID may be combined with the data generated under DI-HFAC-80273 associated with MIL-STD-740-2. (Continued on page 2)

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		N4010

10. PREPARATION INSTRUCTIONS

10.1 General test plan requirements. The plan shall contain the following information:

10.1.1 Test and equipment information.

a. Equipment to be evaluated, including contract number.
b. Ship names and hull number, if known, for which the unit is being acquired.
c. Evaluation measurements to be performed.
d. Evaluating activity and test site.
e. Test date if established.
f. Equipment specification acceptance criteria.
g. Driven and driver unit manufacturers.
h. Driver unit horsepower(HP).
i. Rated electrical characteristics, including volts, amps, phase.
j. Rated fluid media parameters (flow rates, temperatures, pressures, etc.).
k. Equipment rated operational speeds (driver and driven unit revolution per minute(r/min), etc.).
l. Expected equipment noise and vibration sources and their frequencies of excitation (rotationals, gear tooth contact, blade rate, etc.).
m. Resilient mount manufacturer and model number.
n. Dates of manufacture of resilient mounts to be used.
o. Number of resilient mounts to be used.
p. If the use of non-Navy standard resilient mounts has been approved, provide the characteristics including load rating in pounds and resonant frequency data, of the mounts to be used.

(Continued on page 2)

11 DISTRIBUTION STATEMENT

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7. APPLICATION/INTERRELATIONSHIP (Cont'd)

7.4 This DID supersedes UDI-T-23756.

10. PREPARATION INSTRUCTIONS (Cont'd)

q. Planned equipment load during test (full, half, etc.).
r. Planned operating speeds during test.
s. Planned throttling conditions, if any.
t. Planned operating suction and discharge fluid media parameters during test (flow rates, pressures and temperatures, etc.).
u. Planned electrical inputs, outputs or both during test.
v. Block diagrams and identification of make and model of measurement and analysis instrumentation to be used.
w. Block diagrams and identificatioon of make and model of system calibration instrumentation to be used.
x. Place and date of latest laboratory calibration of measurement instruments, system calibration instruments (including field calibrators), and analysis instruments which will be used.

10.1.2 Drawings and sketches to show assembly information, equipment details and the test set up.

a. Overall unit dimensions and orientation (vertical, horizontal, etc.) to be used during test.
b. Identification of major componenents.
c. Interface information.
d. Mounting locations and dimensions.
e. Weights and center of gravity for major componets.
f. Resilient mount orientation (vertical, horizontal, angle), approximate load on each mount, attachment locations, overall unit weight and center of gravity.
g. Pedestal or subbase dimensions, materials and weight, when applicable.
h. Dimensions, materials and weight of the mounting fixture if a mounting fixture other than the standard fixture is to be used.
i. Planned arrangement and description of flexible pipes and ducts, if to be used.
j. Planned arrangement of test piping and fittings.
k. Planned arrangement of electrical power leads.
l. Planned location and description of test instrumentation used to measure r/min, flow rates, total discharge head, etc.
m. Planned locations of resilient mounts and pedestals or fixtures that support the equipment.
n. Approximate locations and orientation of each microphone relative to equipment features.
o. Locations of equipment air inlet and discharge openings.
p. Description of acoustical center of sound source, assumed or measured.
q. Acoustical environment (indoor).

(1) Description of planned location and orientation of test unit relative to floor, walls, ceiling, and contents of test room.
(2) Description of planned test location (anechoic, reverberant, etc.), dimensions of the test room or chamber description of the physical treatment on walls, ceiling and floor.

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10. PREPARATION INSTRUCTIONS (Cont'd)

r. Acoustical environment (outdoor).

(1) Description of planned location and orientation of test unit relative to buildings, structures or other reflecting surfaces or objects.
(2) Description of the planned ground surface, and wind direction and speed.

10.1.3 Test data formats.

a. Tabular or graphic formats to be used in tabulating and recording airborne data.

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